H

TAMPINES JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
[mark scheme]

CANDIDATE NAME							
CIVICS GROUP	1	2				TUTOR NAME	

Biology

8875/ 02

Paper 2

Wednesday, 2 September 2013

2 hours

Candidates answer on the Question Paper

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination:

1. Fasten all answer scripts for Section C securely;
2. Hand in the OAS, Paper 2 and the answer scripts for Section C **separately.**

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use		
Section A		
1	10	
2	10	
3	17	
4	11	
Section B		
5 OR	20	
6	20	
Total	60	

This document consists of **23** printed pages and **1** blank page

SECTION A - STRUCTURED QUESTIONS (80 MARKS)

Answer **all** the questions in this section in the space provided.

QUESTION 1

Fig. **1.1** is a diagram showing part of a bacterial chromosome with a single origin of replication and two replication forks which proceed in opposite directions until replication of the entire bacterial chromosome is complete.

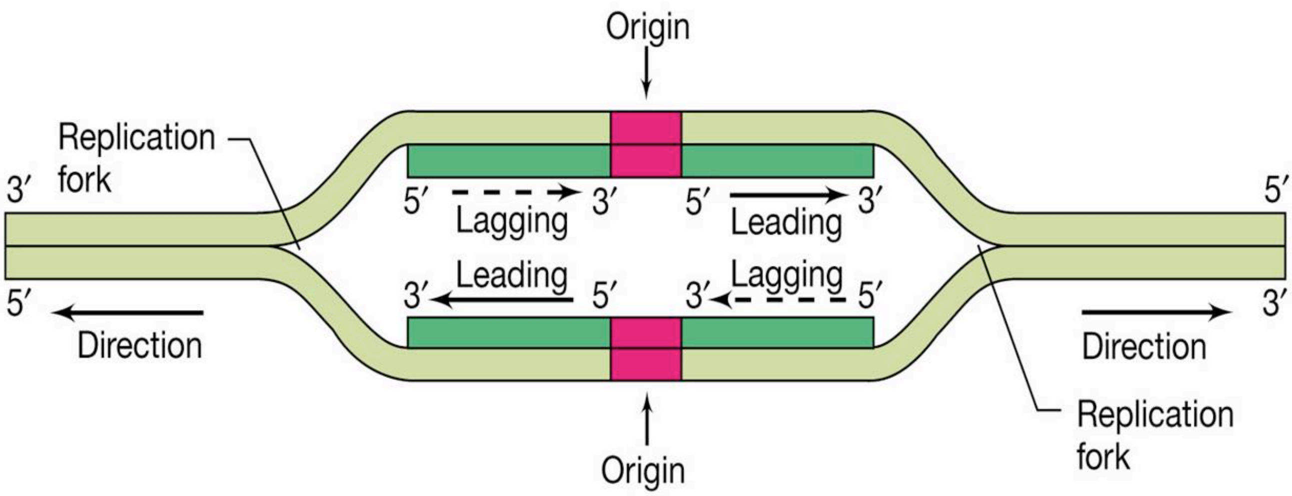


Fig. 1.1

(a) With reference to **Fig. 1.1**, explain why this replication is said to be semi-conservative. [3]

- DNA unzips / DNA strands separate / ref. to H bonds break ;
- Both strands / each strand acts as a template ;
- Each new molecule of DNA contains one new and one old / original strand of DNA

[1 mark each]

(b) With reference to **Fig. 1.1**, replication only occurs in a 5' to 3' direction, explain why this is so. [2]

- structure of active site of DNA polymerase is specific [1]
- allows recognition of a 3' OH from a nucleotide to initiate the replication process ; [1/2]
- active site fits nucleotide in a particular orientation (5' forms phosphodiester bond with 3') [1/2]

The Hayflick limit is the number of times a normal human cell population will divide until cell division stops. Empirical evidence shows that the telomeres, DNA regions that do not code for proteins, associated with each cell's DNA will get slightly shorter with each new cell division until they shorten to a critical length. The Hayflick limit has been found to correlate with the length of the telomere region at the end of a strand of DNA. **Fig. 1.2** shows the position of telomeres in a DNA double helix.

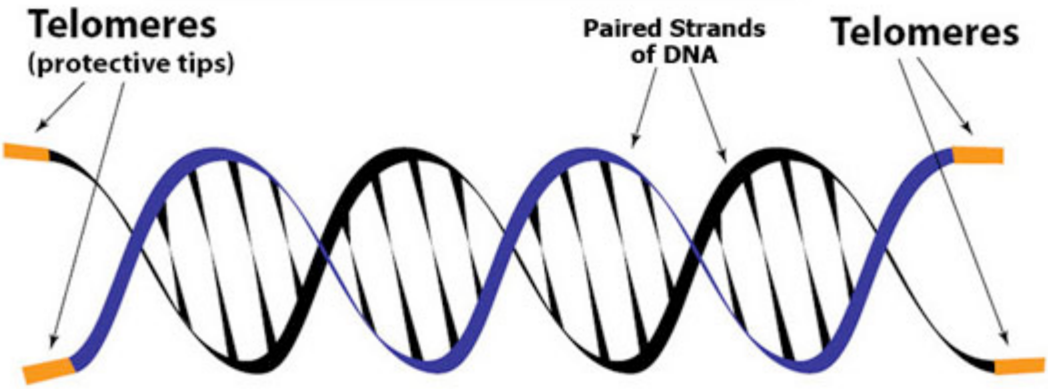


Fig. 1.2

(c) Suggest the role of telomeres. [2]

- To provide stability to the chromosome by rendering chromosome ends generally inert in interactions with other chromosome ends. (i.e. they keep the ends of the various chromosomes in the cell from accidentally becoming attached to each other.)
- The telomere regions acts as a buffer of genes near the ends of chromosomes by allowing for the shortening of chromosome ends, which necessarily occurs during replication.
- Telomere shortening in humans can induce senescence, which blocks cell division. This mechanism appears to prevent genomic instability and development of cancer in human aged cells by limiting the number of cell divisions.

[1 mark each]

In stem cells, the telomeres are protected from shortening with the help of telomerase. Telomerase is an enzyme that lengthens the telomeres. Having lengthened telomeres means that the cell can undergo extensive cellular division without shortening the telomeres to critical lengths.

(d) Some stell cells, such as the zygote is said to be totipotent.

State what it is meant by Totipotent. [2]

- Totipotency is the ability of a single cell to divide indefinitely
- and produce all of the differentiated cells in an organism.

(e) Suggest how having the activity of Telomerase in stem cells allows them to divide indefinitely. [1]

- Telomerase will lengthen the telomeres to prevent its shortening
- Allow cells to divide continuously without having its telomere shorten.

[Total: 10]

QUESTION 2

Carbon is essential to life. All of our molecular machines are built around a central scaffolding of organic carbon. Unfortunately, carbon in the earth and atmosphere is locked in highly oxidized forms, such as carbonate minerals and carbon dioxide gas. In order to be useful, this oxidized carbon must be "fixed" into more organic forms, rich in carbon-carbon bonds and decorated with hydrogen atoms. Powered by the energy of sunlight, plants perform this central task of carbon fixation.

Inside plant cells, the enzyme ribulose bisphosphate carboxylase (rubisco) forms the bridge between life and the lifeless, creating organic carbon from the inorganic carbon dioxide in the air. **Fig. 2.1** shows the structure of ribulose bisphosphate carboxylase (rubisco).

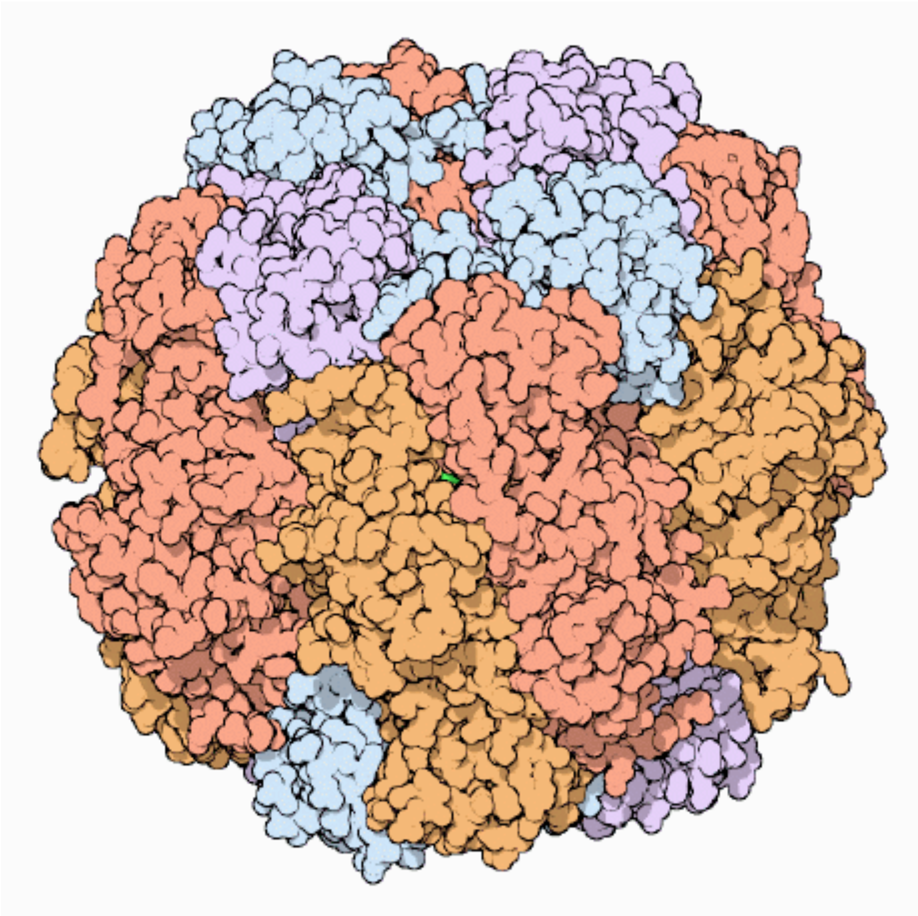


Fig. 2.1

(a) Rubisco is an enzyme made of sixteen polypeptide chains.

Describe how the structure of rubisco is maintained. [2]

- The primary sequence of the 16 polypeptide chains formed by having peptides held between the A.As
- The folding of the individual polypeptides to form secondary structures (Alpha helices and Beta-pleated sheets) and maintained by H bonds between the imino and carbonyl group.
- These secondary structures will form tertiary structures with further compacting and folding and these are held by H2ID interactions
- These tertiary structures will associate with each other to form rubisco with each tertiary structure held by the same interactions as well.

In spite of its central role, rubisco is remarkably inefficient. Typical enzymes can process a thousand molecules per second, but rubisco fixes only about three carbon dioxide molecules per second. **Fig. 2.2** shows the relationship between CO_2 assimilation rate and increasing light intensity in a plant, when carbon dioxide is present in excess.

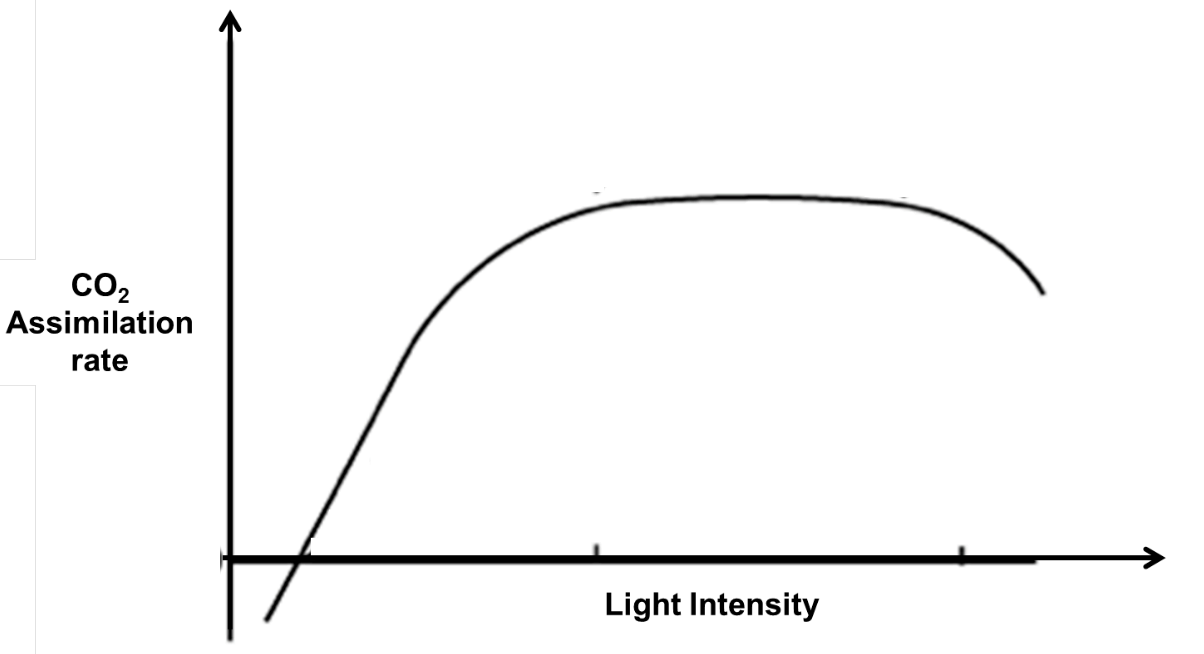


Fig. 2.2

(b) Explain how light energy assists in the process of carbon fixation in plants. [3]

- Light energy is harvested by the antenna complex and causes excitation of electrons in the photosystem I and II
- Put through the process of photophosphorylation
- The excitation of electron of the PS I and II is captured by the primary electron acceptors of each PS
- This electron from PSII is passed down an electron transport chain to generate ATP through chemosis
- The ATP formed will be used during the Calvin cycle during regeneration of RuBP
- In a second ETC, NADP is reduced by NADP reductase to form reduced NAD -> needed for Calvin cycle as a reducing power.

(c) Explain why the CO_2 assimilation rate levels off at higher light intensity. [3]

- CO_2 assimilation rate levels off at high light intensity →maximum rate as all photosystems are saturated with light energy (limited amount of chlorophyll in the photosystems)

- Maximum rate of photosynthesis occurs at high LI hence amount of CO_2 assimilation now remained constant due to the constant (maximum) amount of NADPH produced;

- Light is no longer limiting factor, but other factor e.g. rate of carbon fixation is.

Rubisco also shows an embarrassing lack of specificity. Unfortunately, oxygen molecules and carbon dioxide molecules are similar in shape and chemical properties. In rubisco, an oxygen molecule can bind comfortably in the active site designed to bind carbon dioxide. Rubisco then attaches the oxygen to the sugar chain, forming a faulty oxygenated product.

(c) With reference to **Fig 2.2**, suggest why the CO_2 assimilation rate is declining as the light intensity increases higher. [2]

- As light intensity gets higher, the rate of production of oxygen from photolysis of water to replace the lost of electrons in PS II may increase.
- Carbon dioxide would need time to diffuse into the plant cell through the stomata
- Oxygen produced within the plant cell will be greater than amount of carbon dioxide present
- Oxygen will bind rubisco preferentially and thereby leading to a decline in CO_2 assimilation

[Total: 10]

QUESTION 3

Fig. 3.1 shows Honeycreepers of the Family Drepanididae from the Hawaiian Islands. These Honeycreepers reside on the various Hawaiian Islands. Through a series of studies, these Honeycreepers are found to be evolved from an ancestral Honeycreeper which resided originally on, Kauai, one of the Hawaiian Islands.

Fig. 7.1

(a) Explain how natural selection could have caused the evolution of these various species of Honeycreepers. [4]

Mutation

- Spontaneous mutation or inherited variation in the genotype of these organisms provided genetic variation of each organism

Natural Selection

- Difference in selection pressure (e.g. food type) due to the different habitat that these forms are found.
- There would be one phenotype that would be at a selective advantage in the environment as it is better adapted to the living conditions.
- They will survive, reproduce and pass on their genes to offsprings.

Our understanding of the evolutionary relationship of species has been greatly improved through the use of molecular techniques.

Many evolutionary studies use mitochondrial DNA rather than nuclear DNA.

(b) Explain why mitochondrial DNA is used to determine phylogeny instead of using nuclear DNA. [2]

- Unlike nuclear DNA, whose genes are rearranged in the process of recombination, there is usually no change in mtDNA from parent to offspring.
- the mutation rate of mtDNA is higher than that of nuclear DNA and is easily measured and compared allows mtDNA to be a powerful tool for tracking the ancestry of many species back hundreds of generations
- AVP: Common in all species; valid basis for comparison between distantly related species;

A new Honeycreeper (**Fig. 3.2**) has been found on one of the Hawaiian Islands. Based on its structural features, Scientists are speculating that this new found Honeycreeper is also a descendant of the ancestral Honeycreeper found on Kauai.

Fig. 3.2

(d) Suggest why it is not advisable to determine the evolutionary relationship of this new found Honeycreeper to the rest of them from the morphological features. [2]

- Basing on morphological features is subjective and non-quantifiable.
- Similarities in morphological structures could be due to convergent evolution

---

(c) Explain how you could use molecular homology to show that this new Honeycreeper evolved from the original ancestor Honeycreeper. [2]

- For basis of valid comparison the sequence of similar genes (for example, cytochrome c oxidase gene) are compared amongst the Honeycreepers
- Or sequence of amino acids in the polypeptide of a named protein.
- Degree of similarity in sequences/ degree which sequences of gene are conserved between species are interpreted as DNA evidence for evolution.
- Generally, a higher degree which sequences are being conserved a higher degree of relatedness different species are to one another and vice versa.

[Total: 10]

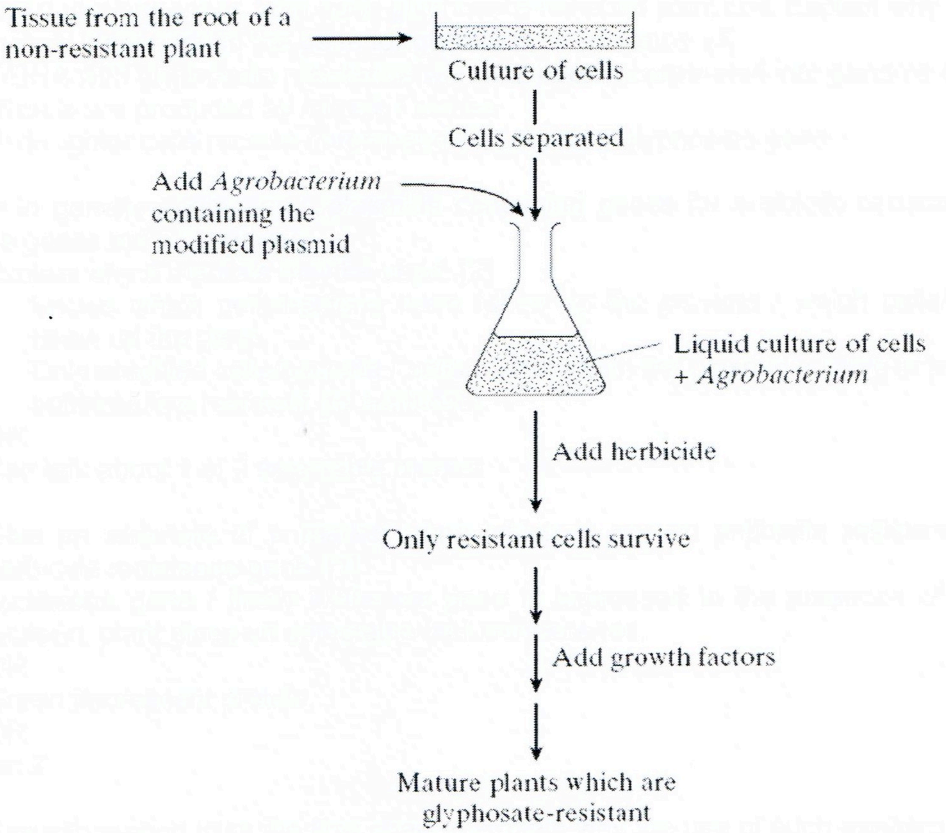
QUESTION 4

Read the following passage.

Genetic engineering is a technique for improving crop plants. It is a process by which a single new gene can be added to the genes already present in the plants. One way of doing this is to use a natural genetic engineer, a soil microorganism called *Agrobacterium*. This bacterium possesses a plasmid which can be modified in the laboratory so that it becomes the carrier of new genetic information.

For example, a gene coding for the ability to break down the herbicide glyphosate can be introduced with the assistance of the enzymes restriction endonuclease and ligase. Once inside the crop plant, the gene for glyphosate breakdown makes the plant resistant to the effects of this herbicide. All surrounding weeds are destroyed when sprayed with glyphosate.

The flow-chart shows how the plasmid carrying this gene can be used to produce glyphosate-resistant plants.



Use the information from the passage and your own knowledge to answer the following questions.

(a) One mature plant grows from each glyphosate-resistant plant cell. Explain why all the cells of the mature plant contain the gene for glyphosate-resistance. [1]

The cells are <u>genetically identical / clone</u> of the glyphosate cells due to <u>mitotic division</u> of the cell.

(b) Outline the production of the genetically modified plasmid. [3]

Restriction enzymes to cut plasmid and gene of interest

To produce sticky ends

That can anneal via H bonds between complementary bases

DNA ligase

Formation of phosphodiester bonds

To form genetically modified plasmid with gene of interest

(c) Suggest and explain **one** possible benefit and problem associated with the use of herbicides together with genetically modified, herbicide-resistant crop plants. [1]

- **The creation of superweeds with horizontal gene transfer**
- **The production of toxins by the gene that can cause allergy in human beings**

Another technique required in plant biotechnology is plant tissue culture. Various organs or tissues of a plant may be used in tissue culture to produce a large number of genetically identical plants.

In the process of anther culture, haploid plantlets are produced from immature anthers, from which mature fertile homozygous plants are obtained. The process is shown diagrammatically in the figure below.

Unopened flower bud
↓
Anthers removed and placed on culture medium **1**
↓
Mitosis of immature pollen cells to give haploid callus
↓
Pieces of callus removed and placed on culture medium **2**
↓
Haploid plantlets
↓
Treated with colchicines to double the chromosome number
↓
Mature fertile homozygous plants

(d) With reference to the figure,

(i) explain what is meant by haploid callus. [2]

- **The cells are genetically identical / undifferentiated glyphosate cells**
- **1 set of chromosomes**

(ii) state one precaution that would be taken when transferring material onto culture medium **1** or **2**. [1]

Ensure that the environment for the transfer is aseptic to prevent contamination.

(iii) Suggest why variation may be generated in plantlets derived from the same callus. [1]
It can be due to <u>chromosomal mutations</u>

(iv) Suggest how colchicine could bring about the doubling of chromosome number in haploid cells. [1]

It may prevent the <u>sister chromatids</u> from separating during <u>anaphase/</u>

<u>Non-disjunction</u>

SECTION B (20 MARKS)

Answer **one** question.
Write your answers on the separate writing paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

QUESTION 5

(a) Compare anaerobic respiration and aerobic respiration. [6]

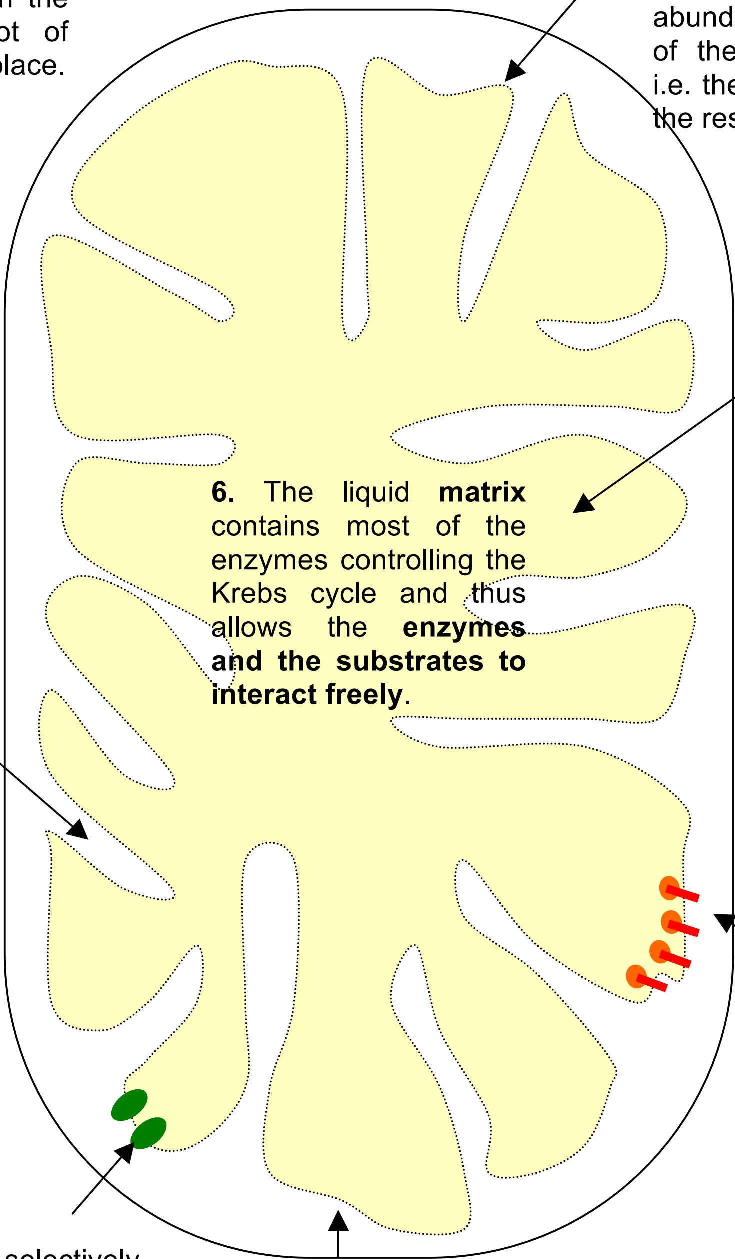
Max 3 for differences	**Anaerobic Respiration**	**Aerobic Respiration**
Definition/ Components	Glycolysis + Fermentation	Glycolysis + Link Reaction + Krebs Cycle + Oxidative phosphorylation
Oxygen requirement	Absence of Oxygen	Presence of Oxygen
ATP production	2 ATP	38 ATP
Electron carriers involved	NAD only	NAD and FAD
Location	Cytoplasm only	Cytoplasm and Mitochondria
Similarities [3 marks]		
• Starting materials is glucose • Both has glycolysis as a step in the process of oxidation of glucose • Both involves the use of electron carriers (NAD or FAD) • Carbon dioxide gas is produced in both processes		

(b) Describe the main features of oxidative phosphorylation. [8]

1) Occurs in the inner mitochondrial membrane
2) Synthesis of ATP occurs as electrons are transferred from reduced NAD and reduced FAD generated from the Krebs cycle, link reaction or glycolysis
3) Down the Electron Transport Chain (ETC) found on the inner mitochondrial membrane
4) The energy released from the electron transport chain is used to actively pump protons (H^+) from the matrix to the intermembrane space of the mitochondria.
5) This thus creates a proton gradient across the inner membrane. The proton gradient serves as a source of potential energy for the synthesis of ATP.
6) The stalked particles on the inner mitochondrial membrane consist of hydrophilic protein channels called F_0 proteins and the enzyme ATP synthase.
7) As H^+ diffuses through the F_0 proteins into the matrix of the mitochondria, the energy released by the protons is used by ATP synthase to attach an inorganic phosphate to ADP, thus forming ATP. The electrochemical gradient is said to be coupled to ATP synthesis.
8) Oxidation of one molecule of NADH yields 3 ATP while oxidation of one molecule of $FADH_2$ yields 2 ATP by oxidative phosphorylation.
9) The electrons are passed down the electron transport chain. O_2 the final electron acceptor is reduced to H_2O.
10) OP allows regeneration of NAD and FAD for the use the other components of respiration.

(c) Discuss how the mitochondrion is adapted for carrying out respiration. [6]

Fig. 9 Adaptation of mitochondrion to aerobic respiration



8. Mitochondria can **move** to areas in the cell where a lot of activity is taking place.

1. The **cristae** of the inner membrane effectively **increase** its **surface area** to provide abundant space for the layout of the multi-enzyme systems. i.e. the sequence of carriers for the respiratory chain.

2. The immediate **bathing of the cristae by the matrix** and the great surface area provided by the cristae allow the carriers **greater access** to the substrates found in the matrix.

6. The liquid **matrix** contains most of the enzymes controlling the Krebs cycle and thus allows the **enzymes and the substrates to interact freely**.

7. Mitochondria in **active** cells are **larger** and contain **more cristae**.

3. Elementary particles associated with ATP synthesis are found on the **matrix side** of the inner membrane

5. The selectively permeable membrane contains translocase enzymes for the **active transport of ATP and ADP through it.**

4. The **inner membrane is impermeable to protons** moving in from the matrix. A proton pump in the inner membrane pumps protons from matrix into intermembrane space.

The **outer membrane is freely permeable to protons** and hence allows protons to move out of the mitochondria. Protons accumulate outside the mitochondria and create a pH gradient. The pH gradient allows protons to diffuse back into the mitochondrion in which ATP is synthesized during the process.

6(a) Describe PCR and compare and contrast with replication process. [6]

1st step: One to several minutes at 94-96 degrees C, during which the DNA is denatured into single strands by breaking of hydrogen bonds.

2nd step: One to several minutes at 50-65 degrees C, during which the primers hybridize or "anneal" (by way of hydrogen bonds) to their complementary sequences on either side of the target sequence.

3rd step: One to several minutes at 72 degrees C, during which the Taq polymerase binds and extends a complementary DNA strand from each primer, using the single stranded DNA as a template.

	Replication	**PCR**
primer	RNA primers	DNA primers
enzyme	DNA polymerase III & I	Taq DNA polymerase
proofreading	yes	no
Temperature dependent	no	yes

(b) Outline the process of gel electrophoresis and explain how it is able to separate fragments of DNA. [9]

Agarose gel electrophoresis is a method for separating and visualizing DNA fragments produced by restriction digestion of DNA.

The fragments are separated by <u>charge and size</u> by forcing them to move through a agarose gel matrix which is subjected to an electric field.

Principles of agarose gel electrophoresis

- Biological molecules such as DNA, RNA and proteins are electrically charged particles at a given pH.
    - DNA is <u>negatively charged</u> due to the phosphate groups of the sugar-phosphate backbone.
    - If the sample contains DNA fragments of different sizes, they can be separated based on their sizes when they are subjected to an electric field.
    - Smaller fragments will move faster then the larger fragments.

Step 1: Casting of the agarose gel

- Agarose is a polysaccharide derived from seaweed that forms a gel when dissolved in an aqueous solution.
- Agarose solution will be poured into casting tray and allowed to solidify.
- With the help of the comb, small indentations called wells are formed at one end of the gel, where DNA samples can be loaded.
- Each well corresponds to one lane.

Step 2: Loading of sample

- DNA samples are first mixed with a loading dye which increases its density and helps the DNA to sink to the bottom of the well.

- ➢ Loading dye also helps to monitor the progress of the electrophoresis as DNA is invisible.
- ➢ Markers are usually run in the 1st lane. Markers are mixtures of DNA fragments of known sizes. By comparing the markers' fragment against the unknown fragments, we can estimate the sizes of the unknown fragments.
- ➢ Samples are loaded into wells with the help of a micropipette.

Step 3: Running of gel

- ➢ Current is switched on.
- ➢ Negatively charged DNA fragments will migrate out of the well and are attracted to the positive electrode.
- ➢ Separation of the fragments will be based on their sizes and can be seen as discrete bands on the gel.

Step 4: Visualization

- ➢ Before the band reaches the end of the gel, current will be switched off.
- ➢ Gel is then stained with DNA binding dye (ethidium bromide which is carcinogenic).
- ➢ Separated DNA fragments can been clearly under the UV light.

(c) State the uses of RFLP analysis. [3]

- **Genomic mapping in terms of linkage mapping**
- **Disease detection, e.g sickle cell anaemia**
- **DNA fingerprinting**

End of Paper

BLANK PAGE